ROS2 for ATOSFleetManagement Cheat Sheet

1) Source environment
```bash
source /opt/ros/humble/setup.bash
source /home/sepast/atos_ws/install/setup.bash
```

2) Build ATOSFleetManagement packages
```bash
cd /home/sepast/atos_ws
colcon build --packages-select atos atos_gui --symlink-install
source install/setup.bash
```

3) Start ATOSFleetManagement (normal)
```bash
ros2 launch atos launch_atosfleetmanagement.py insecure:=True
```

4) Start ATOSFleetManagement with 3 simulators
```bash
ros2 launch atos launch_atosfleetmanagement.py insecure:=True with_truck_simulator:=True
```

Default simulator setup in launch
- `L5S-TRUCK-SIM-1`: `start_index=0`, `target_speed_kmh=80`, `ignore_warning_speed_commands=True`
- `L5S-TRUCK-SIM-2`: `start_index=250`, `target_speed_kmh=40`
- `L5S-TRUCK-SIM-3`: `start_index=500`, `target_speed_kmh=40`

5) Open GUI
- URL: `http://localhost:3000`
- Go to tab: `RuralRoad Map`
- UI shows live trucks + `Distance To Next Truck Ahead`

6) Check running nodes
```bash
ros2 node list
```

7) Check key topics
```bash
ros2 topic list | rg "truck_objects|speed_command"
```

8) Watch live truck states for GUI
```bash
ros2 topic echo /atos/truck_objects/state
```

9) Watch control speed commands
```bash
```

```
ros2 topic echo /atos/truck_objects/speed_command
```

10) TruckObjectControl COT interfaces
- ROS topic input: `/atos/truck_objects/cot` (placeholder format)
- TCP input: `0.0.0.0:8114` listener in TruckObjectControl
- Truck clients connect to: `127.0.0.1:8114` (same machine) or `<host-ip>:8114`

11) Placeholder ROS COT test (manual)
```bash
ros2 topic pub /atos/truck_objects/cot std_msgs/msg/String "data: 'id=truck_1;distance_m=1000;tcp_connected=1;lat=57.78357;lon=12.76389;speed_kmh=10;course_deg=150'" -1
```

12) Start one simulator manually (custom test)
```bash
ros2 run atos atos_truck_simulator --ros-args \
  -p uid:=L5S-TRUCK-UBUNTU \
  -p start_index:=300 \
  -p target_speed_kmh:=35.0 \
  -p acceleration_mps2:=0.6 \
  -p tcp_host:=127.0.0.1 \
  -p tcp_port:=8114
```

13) Simulator option: ignore first-limit slowdown
Use this only on selected trucks:
```bash
-p ignore_warning_speed_commands:=true
```
Behavior:
- Ignores non-zero warning command (for example `30 km/h` at `< 400 m`)
- Still obeys stop command (`0 km/h` at `< 200 m`)

14) Fleet control thresholds
- Gap `< 400 m` => publish `target_speed_kmh=30`
- Gap `< 200 m` => publish `target_speed_kmh=0`

15) Stop everything
- Press `Ctrl+C` in launch terminal